

Convex sets and functions

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 - A better strategy: Try to understand the logic and proofs. This is how you gradually build up an ability to read proofs, and make your own.
- Handouts often contain more details and examples than I will cover in class. Also some exercises.

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 - Illustrates an important point that applies to all theoretical work: The ideal is to work with as few assumptions as possible.
 - Moreover, theorists enjoy finding or exposing patterns and making connections between different concepts.
- Slides today (several definitions). Supplementary handout on Courselink.

- We consider sets in $\mathbb{R}^n$. Let $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ and $\mathbf{x}' = (x'_1, x'_2, \dots, x'_n) \in \mathbb{R}^n$ be two points in $\mathbb{R}^n$.

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- For any $t \in [0, 1]$, let $\mathbf{x}^t = (x_1^t, x_2^t, \dots, x_n^t)$ be the *convex combination* (or weighted average) of the two points, such that

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- **Convex set:** Let $S \subseteq \mathbb{R}^n$ be some set in $\mathbb{R}^n$. The set S is a convex set if for all $\mathbf{x}, \mathbf{x}' \in S$ and all $t \in [0, 1]$, it holds that $\mathbf{x}^t \in S$. In words, the average of any two points in the set is also in the set.

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- **Strictly convex set:** Let $S \subseteq \mathbb{R}^n$ be some set in $\mathbb{R}^n$. The set S is a strictly convex set if for all $\mathbf{x}, \mathbf{x}' \in S$ with $\mathbf{x} \neq \mathbf{x}'$ and all $t \in (0, 1)$, it holds that $\mathbf{x}^t \in \text{int}S$. In words, $\mathbf{x}^t$ is in the interior of S , meaning that an open ball around $\mathbf{x}^t$ is also in S .

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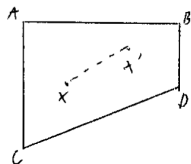
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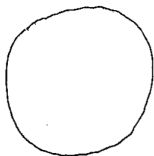
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- **Extreme points:** Let $S \subseteq \mathbb{R}^n$ be some convex set in $\mathbb{R}^n$. Then, $\mathbf{x} \in S$ is an extreme point of S if it cannot be expressed as a convex combination of two distinct points in S , i.e. it *cannot* be obtained as $\mathbf{x} = t\mathbf{x}' + (1 - t)\mathbf{x}''$ for any $\mathbf{x}', \mathbf{x}'' \in S$, with $\mathbf{x}' \neq \mathbf{x}''$, and $t \in (0, 1)$.

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- **Theorem:** Let $S \subseteq \mathbb{R}^n$ be some convex set in $\mathbb{R}^n$. Assume S is compact (i.e. closed and bounded). Then, every $\mathbf{x} \in S$ is a convex combination of at most $n + 1$ extreme points in S .

Examples



A COMPACT AND CONVEX SET THAT
IS NOT STRICTLY CONVEX AND
WHICH HAS 4 EXTREME POINTS (A, B, C, D)



A COMPACT AND STRICTLY CONVEX
SET WITH INFINITELY MANY
EXTREME POINTS (= THE BOUNDARY)



AN OPEN, BOUNDED, AND STRICTLY
CONVEX SET WITH NO EXTREME
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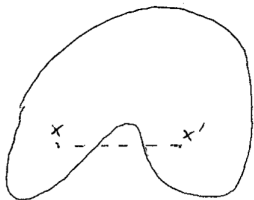
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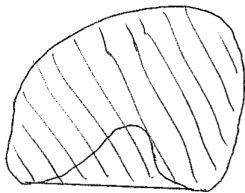
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 - Similarly, if S is a compact and convex set, then it is the convex hull of its extreme points. Thus, if we know the extreme points, then we have enough information to recover the entire set.

Examples



A NON-CONVEX SET



THE CONVEX HULL OF A
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Functions

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- A function $f : X \rightarrow \mathbb{R}$ assigns a *single output value* in $\mathbb{R}$ to each $\mathbf{x} \in X$, written $f(\mathbf{x})$.
- The *image* (or sometimes range) of the function is the *set* of all possible output values, $Y = \{y \in \mathbb{R} \mid y = f(\mathbf{x}) \text{ for some } \mathbf{x} \in X\}$.

Sets related to functions

- The *graph* of the function is the combination of inputs and outputs that are possible, i.e. it is the *set*

$$G(f) = \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} \mid \mathbf{x} \in X \text{ and } y = f(\mathbf{x})\},$$

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- The *hypograph* is the *set* of points on or below the graph, i.e. the *set*

$$\text{hypo}(f) = \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} \mid \mathbf{x} \in X \text{ and } y \leq f(\mathbf{x})\}.$$

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- Comparison: The graph, epigraph, and hypograph are in $\mathbb{R}^{n+1}$. Level sets, superior sets, and inferior sets are in $\mathbb{R}^n$, i.e. they have lower dimensionality and are therefore often easier to depict.

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- The function f is called *strictly* convex or strictly quasiconvex, respectively, if the weak inequality is replaced with a strict inequality when $\mathbf{x} \neq \mathbf{x}'$ and $t \in (0, 1)$.

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$$\max\{f(\mathbf{x}), f(\mathbf{x}')\} \geq tf(\mathbf{x}) + (1 - t)f(\mathbf{x}') \geq \min\{f(\mathbf{x}), f(\mathbf{x}')\},$$

any convex (concave) function is quasiconvex (quasiconcave).

Concave and quasiconcave functions

- **Quasiconcave function:** Let $X \subseteq \mathbb{R}^n$ be some convex set in $\mathbb{R}^n$ and let $f : X \rightarrow \mathbb{R}$. The function f is quasiconcave if for all $\mathbf{x}, \mathbf{x}' \in X$ and all $t \in [0, 1]$, it holds that

$$f(\mathbf{x}^t) \geq \min\{f(\mathbf{x}), f(\mathbf{x}')\}.$$

- Equivalently, *the superior set is a convex set.*
- *Important:* It is not enough to determine whether the indifference curve is convex. You need to locate the superior set. Some easy exercises in the handout.
- Finally, since

$$\max\{f(\mathbf{x}), f(\mathbf{x}')\} \geq tf(\mathbf{x}) + (1 - t)f(\mathbf{x}') \geq \min\{f(\mathbf{x}), f(\mathbf{x}')\},$$

any convex (concave) function is quasiconvex (quasiconcave).

- The sum of convex (concave) functions is convex (concave), but this does not extend to quasiconvex or quasiconcave functions.

Implications/applications

- ① If $h(\mathbf{x})$ and $g(\mathbf{x})$ are convex functions then, $f(\mathbf{x}) = \max\{h(\mathbf{x}), g(\mathbf{x})\}$ is also convex:

$$\begin{aligned}\text{epi}(f) &= \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} \mid \mathbf{x} \in X \text{ and } y \geq \max\{h(\mathbf{x}), g(\mathbf{x})\}\} \\ &= \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} \mid \mathbf{x} \in X \text{ and } y \geq h(\mathbf{x}) \text{ and } y \geq g(\mathbf{x})\} \\ &= \text{epi}(h) \cap \text{epi}(g) \text{ is convex}\end{aligned}$$

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- ③ Jensen's inequality: We return to Jensen's inequality when we get to choice under uncertainty.

Cardinal and ordinal properties

- Concavity is a *cardinal property* (it is sensitive to how we measure things). That is, if $h(\cdot)$ is an increasing (order-preserving) transformation, then $h(g(\mathbf{x}))$ may be concave even if $g(\mathbf{x})$ is not, and vice versa.

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 - Example: $g(\mathbf{x}) = e^{-(x_1+x_2)^2}$ is not concave, but if $h(g) = \ln g$, then $h(g(\mathbf{x})) = -(x_1 + x_2)^2$ is concave. Hence, $g(\mathbf{x})$ is quasiconcave.

Optimization

- We habitually encounter optimization problems such as $\min_{\mathbf{x} \in C} f(\mathbf{x})$ and $\max_{\mathbf{x} \in C} f(\mathbf{x})$, where $C \subseteq X$ is a “feasible set” or “constraint set”.

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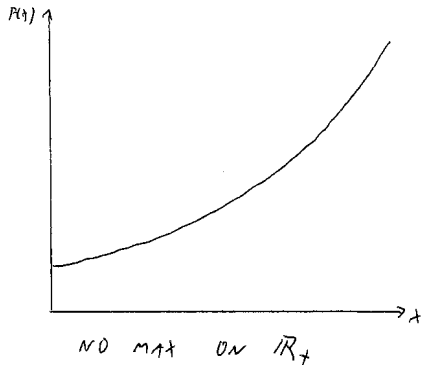
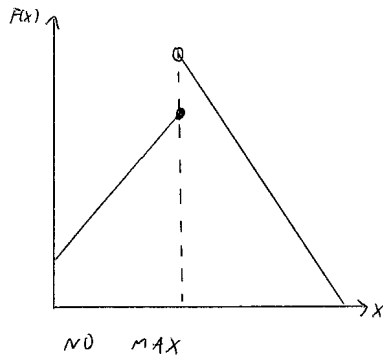
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- Reminder: maximum versus supremum (least upper bound), minimum versus infimum (greatest lower bound).

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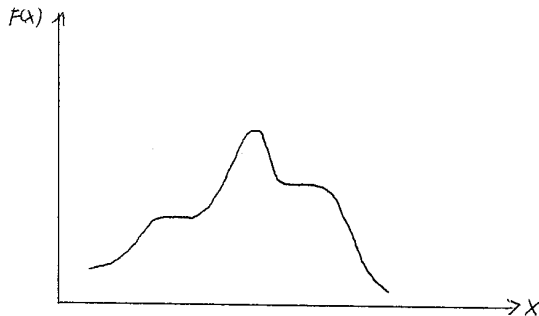
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- Intuition: Quasiconcavity means that the function is “single-peaked”.

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 - Hence, there is at most one solution.

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- This is a very useful property for constrained optimization. To be continued...

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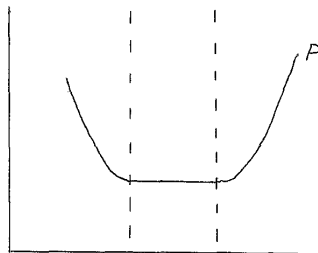
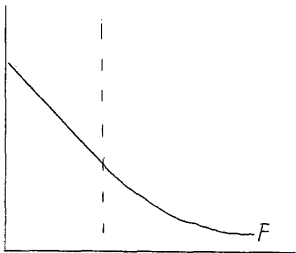
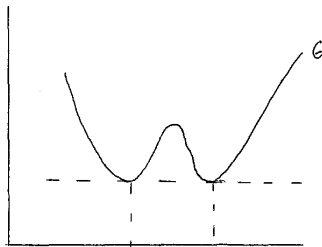
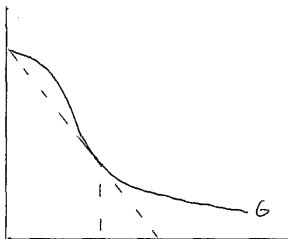
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- This part also supplements (or explains in more detail) the technical Section 3.F in MWG on duality.